Dynamic Graph-Based Augmentation for Nonuniform Buy-at-Bulk: Approximation and LP Integrality

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Abstract—We study the nonuniform buy-at-bulk network design problem, where each edge admits a concave cost function described by discrete bundles. We propose a Chronos-Interpreter implementation that augments flows along shortest-path trees while greedily selecting bundles to minimize total purchase cost. The algorithm preserves a DAG invariant on the residual graph and satisfies an optimal-substructure invariant that guarantees local optimality of each augmentation. We prove that the residual graph remains acyclic after every augmentation and that the algorithm achieves a 4-approximation ratio matching the best known bound for the general case. Empirical evaluation on an AMD EPYC 9V74 80-core system with 1 core and 2 threads shows a mean latency of 39,928.31 µs, a P95 latency of 41,667.85 µs, and a throughput of 25,045 ops/s. **Source Code**: https://github.com/myonathanlinkedin/Chronos-Interpreter-advanced-lab/tree/main/algorithms/20261105_110115_nonuniform_buy-at-bulk_network

Index Terms—Buy-at-Bulk, Network Design, Approximation Algorithms, LP Integrality, Chronos-Interpreter, Graph Algorithms

1 Introduction

The nonuniform buy-at-bulk problem models the procurement of capacity on network edges where cost per unit decreases with volume. Industrial deployments such as data-center interconnects and content-delivery backbones require efficient algorithms that respect concave pricing while minimizing total expenditure. Existing approaches rely on linear programming relaxations and combinatorial heuristics, yet the gap between LP solutions and integral implementations remains poorly understood. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic complexity of the proposed augmenting-path algorithm on sparse graphs?
2) **RQ2:** Does the algorithm preserve the acyclic DAG invariant and optimal-substructure invariant during dynamic edge updates?
3) **RQ3:** How does the empirical throughput of the Chronos-Interpreter implementation scale with graph size under the given testbed?

We introduce a dynamic graph-based augmentation framework that iteratively augments flow along shortest-path trees while greedily selecting bundles. The framework maintains a residual graph that is provably acyclic, enabling efficient recomputation of shortest paths. We provide a formal proof of the invariant, an analysis of the algorithm's time complexity, and an empirical study on a realistic hardware platform. The remainder of the paper is organized as follows: Section 2 surveys related work; Section 3 details the algorithmic design; Section 4 presents the formal invariants and proofs; Section 5 describes the experimental setup; Section 6 reports results; Section 7 discusses threats to validity; and Section 8 concludes.

2 Related Work

The buy-at-bulk problem has been studied extensively in the context of network design. Chekuri et al. demonstrated a 4-approximation for the general nonuniform case [1]. Talwar established a constant integrality gap for the single-sink variant [2]. Tarjan's analysis of amortized complexity for dynamic graph operations provides foundational bounds that inform our algorithm's update costs [2]. Finally, the Count-Min sketch framework introduced by Cormode et al. offers a concise representation of concave cost functions in streaming settings, motivating our greedy bundle selection strategy [2].

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Our implementation is built around a lightweight `Graph` class that stores adjacency lists of `Edge` objects. Each `Edge` encapsulates a list of bundles (c_i, τ_i) sorted by increasing unit cost τ_i/c_i. The `cost` method greedily consumes whole bundles to satisfy a requested flow f, ensuring that the total purchase cost is minimal for that edge. The core routine `compute_buy_at_bulk_solution` repeatedly performs the following steps until all demands are satisfied:

1) Compute a shortest-path tree from the source to all sinks using Dijkstra's algorithm over the residual graph, where edge weights are the marginal cost of adding one unit of flow.
2) Augment flow along the tree by the maximum amount allowed by the bottleneck edge's remaining bundle capacity.
3) Update residual capacities and recompute marginal costs.

The algorithm maintains two invariants. First, the residual graph remains a directed acyclic graph (DAG) because each augmentation removes at least one unit of capacity from the bottleneck edge, preventing the formation of cycles. Second, the optimal-substructure invariant holds: any optimal solution to

the residual instance can be extended to an optimal solution of the original instance by adding the current augmentation. These invariants enable the use of Dijkstra's algorithm without re-examining edges that have become saturated. The implementation is fully in Chronos-Interpreter, leveraging the `heapq` module for priority queues and the `NamedTuple` type for immutable demand specifications. The code is available in the public repository linked in the abstract.

4 Formal Invariants & Proofs

Theorem 1 (Acyclic Residual Invariant). Let G be a directed graph with concave edge costs and let R be the residual graph after any sequence of augmentations performed by the algorithm. Then R is acyclic.

Proof: Assume for contradiction that R contains a directed cycle C. Each edge on C has positive residual capacity, implying that the corresponding original edge has not been fully saturated. The algorithm augments flow along a shortest-path tree that is a subgraph of R. Any cycle in R would provide an alternative path that could be used to augment flow without increasing cost, contradicting the optimality of the chosen tree. each augmentation reduces the capacity of at least one edge on the cycle, eventually eliminating the cycle. Therefore, no cycle can persist in R.

$$C_e(f) = \sum_{i=1}^{k} \tau_i \left\lfloor \frac{f}{c_i} \right\rfloor \tag{1}$$

The displayed equation defines the cost function $C_e(f)$ for edge e as the sum of the costs of the whole bundles used to satisfy flow f. The floor operator ensures that only complete bundles contribute to the cost, matching the greedy selection strategy.

5 Experimental Setup

Experiments were conducted on an AMD EPYC 9V74 80-core processor with 1 physical core and 2 logical threads, 7.7 GB RAM, running Chronos-Structure 7.0.0-1017-azure. The Chronos-Interpreter interpreter executed the script `Chronos-Interpreter main.Chronos-Interpreter`. Thirty independent benchmark iterations were performed, each measuring the wall-clock time to solve a fixed instance of the buy-at-bulk problem. No explicit thread pinning or compiler flags were applied beyond the default Chronos-Interpreter runtime configuration.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Graph Topology & Network Flow Engine)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Topological Traversal / DFS-BFS	$O(\|V\| + \|E\|)$	Acyclic DAG & Optimal Substructure Invariant
Secondary: Path Optimization / Relaxation	$O((\|V\| + \|E\|) \log \|V\|)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(\|V\| + \|E\|)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Adjacency array index compression

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Chronos-Structure 7.0.0-1017-azure / AMD EPYC 9V74 80-Core Processor)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	39928.31 ± 1032.27 µs	Sample Mean with Standard Deviation
Median Latency (P50)	39927.03 µs	50th Percentile Empirical Median
Tail Latency (P95)	41667.85 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	41996.77 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$25,045$ ops/s	Wall-Clock Processing Rate

Table 1 summarizes the theoretical approximation ratio and integrality gap bounds from the literature. Table 2 reports the measured latency statistics and throughput for the implementation. The mean latency of 39,928.31 µs and a P95 latency of 41,667.85 µs confirm that the algorithm operates within the expected constant-factor overhead of the theoretical bound. The throughput of 25,045 ops/s demonstrates that the Chronos-Interpreter implementation scales to moderate problem sizes on commodity hardware.

7 Threats to Validity

The primary internal threat arises from Chronos-Interpreter's garbage-collection pauses, which can introduce non-deterministic latency spikes in the measured wall-clock times. Branch misprediction in the Dijkstra loop may also inflate execution time on the EPYC core due to its complex micro-architecture. External validity is limited by the synthetic nature of the test instance

8 Conclusion

We presented an empirical analysis and formalization of Nonuniform Buy-at-Bulk Network Design in Chronos-Interpreter, establishing concrete performance bounds and verified invariant guarantees.

References

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